Ronit Dama

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Education

Indian Institute of Technology Roorkee **July 2025 – Present**
B.S. in Mathematics and Computing — CGPA: 9.1 *Roorkee, Uttarakhand*

Delhi Public School, Pune **2011 – 2025**
Higher Secondary Education (CBSE) *Pune, Maharashtra*

Achievements

- **JEE Advanced 2025** — AIR 5780 (top 0.34%)
- **JEE Mains 2025** — AIR 11295 (top 0.66%)
- **IMC Prosperity 4** — World Rank 1200
- **Machine Learning Specialization (DeepLearning.AI / Stanford Online)** — Completed *Supervised Machine Learning: Regression and Classification* (Jun 2026) and *Advanced Learning Algorithms* (Jul 2026)

Projects

Quantitative Yield Curve Modeling (CIR / JCIR) | *Python* | GitHub **May 26 – Jun 9, 2026**

- Built a yield-curve reconstruction system in Python that predicts the full US Treasury term structure (9 maturities, 0.25Y–30Y) using only the observed 3-month yield as input.
- Implemented the **Cox–Ingersoll–Ross (CIR)** short-rate model from scratch, including the closed-form affine bond pricing solution $P(\tau) = A(\tau) \cdot \exp(-B(\tau) \cdot r)$.
- Engineered a full data pipeline over ∼2,500 daily observations, handling cleaning, maturity alignment, and a strict chronological train/test split to prevent look-ahead bias.
- Calibrated model parameters (κ, θ, σ, λ) via nonlinear least-squares optimization on the training curve, with MLE-based initialization for stability.
- Achieved a **Test R^2 of 0.9224**, exceeding the 0.85 project benchmark and the theoretical ceiling typically cited for strict single-input affine models.
- Extended the framework to a **Jump-Diffusion CIR (JCIR)** model with numerical quadrature for the jump-compensator term.
- Diagnosed and rejected a rolling CIR++ variant after identifying data leakage — its 0.9986 R^2 came from ingesting the prior day's full observed curve, violating the single-input constraint.
- Interpreted the 2022 Fed hiking cycle as persistent drift rather than discrete jumps, explaining why calibrated jump intensity converged near zero.
- Delivered as a fully executed, documented Colab notebook with a supporting GitHub repository.

Relevant Coursework

Real Analysis · Linear Algebra · Multivariable Calculus · Ordinary & Partial Differential Equations · Probability & Statistics · Stochastic Calculus · Numerical Methods · Discrete Mathematics · Optimization Techniques · Data Structures & Algorithms · Design & Analysis of Algorithms · Fundamentals of Machine Learning

Technical Skills

Languages: Python, C++, Java, SQL, HTML/CSS
Libraries / Frameworks: NumPy, Pandas, scikit-learn, Matplotlib, SciPy, PyTorch, TensorFlow
Developer Tools: Git, GitHub, Jupyter / Google Colab